

Umarfarook Gurramkonda

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SUMMARY

Founding ML engineer with production experience shipping LLM workflows, RAG pipelines, and agentic systems at an early-stage AI startup. Strong Python (FastAPI), hands-on with LangChain, embedding models (BERT-family), multi-LLM orchestration with fallback routing, streaming responses, and API-first integrations. Owns end-to-end delivery from data ingestion and model integration through retrieval tuning, API design, eval harnesses, and containerized deployment on GCP and AWS.

EXPERIENCE

Founding ML Engineer, Hypeon AI

Oct 2025 to Present

- Architected and shipped a production multi-stage LLM orchestration service with hierarchical agentic workflows (routing, intent, retrieval, analysis, composition sub-agents), SSE streaming, and Pydantic-validated structured outputs.
- Designed and deployed a natural-language-to-SQL system over a BigQuery data warehouse with schema introspection, synonym matching, and dry-run cost guardrails; Claude Haiku primary with Gemini fallback routing.
- Built and tuned RAG pipelines using LangChain and BERT-family embedding models (sentence-transformers, all-MiniLM-L6-v2); refined chunking and metadata filtering for retrieval precision.
- Integrated Claude (Sonnet, Haiku), Gemini, and OpenAI behind a unified fallback router with prompt versioning, structured-output validation, and injection guardrails; deployed on GCP Cloud Run with Cloud SQL, Redis, Alembic migrations, Prometheus metrics, and GitHub Actions CI/CD.

Freelance ML / AI Engineer

Oct 2024 to Sep 2025

- Built an AI-powered inventory management system for a retail client: automated invoice generation using LLM-based extraction, real-time stock alerts, and reorder recommendations.
- Implemented stock-level forecasting and demand prediction using Pandas, NumPy, and scikit-learn; surfaced insights through a data-analysis and visualization dashboard.
- Stack: Python, Pandas, NumPy, scikit-learn, OpenAI API, SQL.

Backend Developer Intern, Synclovis Systems

Jun to Sep 2024

- Built RESTful backend services using Node.js, Express, and MySQL for an event-management web application; routing, middleware, and structured data storage.
- Contributed to an internal LLM-based healthcare assistant: integrated RAG retrieval (LangChain + FAISS) over clinical documents and added guardrails to reduce hallucinations.

EDUCATION

B.Tech, Computer Science | K.S.R.M College of Engineering, JNTU Anantapur

2024 (CGPA: 8.14 / 10)

SELECTED PROJECTS

mcp-bigquery-evals: BigQuery MCP Server with Cost Guardrails & Eval Harness

Python, MCP, BigQuery, pytest

- Open-source MCP server (PyPI v0.1.0) exposing 7 read-only BigQuery tools for LLM agents with mandatory dry-run cost caps before every query and 7 stable error codes for agent-side error handling.
- Built a Spider/BIRD-style result-set-equivalence eval harness with a live accuracy badge auto-recomputed per release; bundled sqlite-backed in-memory FakeBigQueryClient for local dev and CI without GCP credentials.

Conversational Research Agent: Multi-stage LLM Orchestration

Python, FastAPI, LangChain

- Multi-stage routing (chitchat, factual, research) with streaming responses, session-based memory, idempotent retries, Pydantic-validated outputs, prompt-injection guardrails, and Prometheus metrics.

AI-Powered Inventory Management Platform

Python, Pandas, scikit-learn, OpenAI

- Automated invoice generation via LLM extraction, real-time stock alerts, demand forecasting with scikit-learn, and a data-visualization dashboard for business insight.

SKILLS

- **Programming:** Python, SQL, JavaScript / TypeScript; Pandas, NumPy, Scikit-learn
- **Deep Learning / NLP:** Neural networks, Transformer architecture, word embeddings, sequence modeling, BERT-family (sentence-transformers), HuggingFace, TF-IDF, sentiment analysis
- **LLMs / GenAI:** LangChain, RAG, agentic AI, multi-stage orchestration, prompt engineering, structured outputs (Pydantic), streaming (SSE); OpenAI (GPT), Claude (Sonnet, Haiku), Gemini
- **Retrieval:** FAISS, semantic search, chunking strategies, metadata filtering
- **Backend:** FastAPI, Flask, REST APIs, async SQLAlchemy
- **Cloud / Data:** GCP (Cloud Run, Cloud SQL, BigQuery, Memorystore), AWS (S3, EC2), PostgreSQL, Redis, Oracle Cloud, Azure
- **DevOps / Tools:** Docker, Git, GitHub Actions CI/CD, Alembic, Prometheus